THE UNITED REPUBLIC OF TANZANIA

PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

(KATAVI, MBEYA, NJOMBE, RUKWA, RUVUMA AND SONGWE)

FORM FOUR ZONAL MOCK II EXAMINATION

032/2A CHEMISTRY 2A

(Actual Practical)

TIME: 2:30 HOURS **YEAR: 2025**

INSTRUCTIONS

1. This paper consists of **two (2)** questions.
2. Answer **all** questions.
3. Each question carries **twenty-five (25)** marks
4. Non programming calculators may be used.
5. All writings must be in **blue** or **black** ink, **except** drawing which must be in pencil.
6. All communication devices and any unauthorized materials are **not** allowed in the examination room.
7. Write your **Examination Number** on every page of your answer sheets provided.
8. You may use the following constants:-

   Atomic masses: H = 1, O = 16, Na= 23, C=12

   1litre = 1dm3 = 1000cm3

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1. You are provided with the following;

AA: A solution made by diluting 50cm^3 of 1.5M oxalic acid o make 5dm^3 of a solution.

BB. A solution containing 0.45gm of impure sodium hydroxide in 0.5dm^3 of a solution.

Phenolphthalein indicator (**P.O.P)** and methyl orange (**M.O**) indicator.

Procedure;

(i) Pipette 20cm^3 or 25cm^3 of solution **BB** and put it in the titrating flask then add two drops of the indicator of your choice as given.

(ii) Titrate solution **AA** in the burette against solution **BB** in the flask until the colour change is observed.

(iii) Record the volume of solution **AA** required to neutralize solution **BB** .

(iv) Repeat the procedure (i) to (iii) to obtain three more readings of solution **AA** and tabulate your data.

Questions;

(a) Calculate the average volume used.

(b) (i) _____cm^3 of solution **AA** required _____cm^3 of solution **BB** for a complete neutralizion reaction

(ii) The colour change was from _____to _____

(c) Give reason for your choice of indicator.

(d) With state symbols write a well-balanced ionic chemical equation.

(e) Calculate the molarity of solution **AA.**

(f) Showing your calculations clearly, calculate the percentage purity of Sodium hydroxide.

(g) How the reaction between **AA** and **BB** can be applied in our daily life (give two points)

(h) Suggest two sources of error in this experiment

2. You are provided with the following

X: 20g of solid sodium thiosulphate ($Na_2S_2O_3$) in 1 litre solution

Y: 2M Hydrochloric acid

Distilled water
Stop watch
Thermometer
Plain paper

Procedure

(i) Draw a letter W on white paper

(ii) Measure 20cm^3 of solution X into 100cm^3 beaker. Add 20cm^3 of distilled water and heat the contents.

(iii) When the temperature reaches 60^{0}C, transfer the beaker on top of a cross and immediately add 5cm^3 of solution Y at the same time start a stop watch.

(iv) Stir the mixture with glass rod, while observing the letter W on top view

(v) Stop the clock immediately as soon as the letter W is obscured and record the time

(vi) Repeat the procedure at a temperature of 50^{0}C, 40^{0}C and 30^{0}C and record the data in a given table below

Experiment	Temperature (^{0}C)	Volume of X (cm^3)	Volume of Y (cm^3)	Volume of water (cm^3)	Time (sec)	1/t (sec$^-$)
1	60	20	5	20		
2	50	20	5	20		
3	40	20	5	20		
4	30	20	5	20		

QUESTIONS

(a) What is the aim of this experiment
(b) Complete the table above
(c) By giving reason(s), identify the experiment in which the reaction was;

(i) Fast
(ii) Slow

(d) With state symbols write the ionic equation for the reaction between X and Y
(e) (i) Draw the graph of temperature (^{0} C)against time
(ii)What conclusion can you draw from the graph? Give reasons for your answer.